

YANNIAN NIU

Phone: (+1) 959-210-4268 ♦ **Email:** yannian.niu@uconn.edu

EDUCATION

University of Connecticut

August 2023 - Present

Ph.D. in Computer Science and Engineering (GPA: 4.00)

Advisor: Dr. Minmei Wang

East China Normal University

September 2020 - May 2023

Master of Science in Software Engineering

East China Normal University

September 2016 - May 2020

Bachelor of Engineering in Software Engineering

PUBLICATIONS

(VLDB 2026) Niu Y, Han S, Wang M. STEM²: A Fast and Space-efficient Data Structure for Exact Multi-Set Membership Queries. 52nd International Conference on Very Large Data Bases.

(NFV-SDN 2025) Niu Y, Wang M. MagicBox: Towards Consolidated Algorithms of Co-existing Network Functions on Heterogeneous Devices. IEEE Conference on Network Function Virtualization and Software-Defined Networking.

(IEEE ToN) Niu Y, Wang M. MagicBox: Towards Consolidated Algorithms of Co-existing Network Functions on Heterogeneous Devices. IEEE Transactions on Networking (extended version). **(Under review)**

(SIGMOD 2027) Niu Y, Wang M. STEM³: From Set Intersection to Point Query for Scalable Metadata Filtering. ACM International Conference on Management of Data. **(Under review)**

RESEARCH EXPERIENCE

Line-Rate Exact Membership Query on Programmable Switch

June 2026 - Present

- Deployed the STEM² multi-set membership query algorithm onto programmable switches via P4, enabling exact per-packet flow classification and multi-tenant ACL enforcement at line rate (Tbps) — use cases that traditional CPU-based solutions cannot sustain under modern data-center traffic volumes.
- Implemented a full control/data-plane co-design on Intel Tofino switch, resolving hardware-specific constraints including pipeline stage budgeting, stateful ALU register packing, and atomic control-plane update protocols for zero-downtime rule changes.

Efficient Metadata Indexing Solution by using an Efficient Multi-Set Multi-Membership Query Algorithm

January 2026 - Present

- Built an exact multi-set multi-membership index using a novel ternary-separator partition tree and an overlap-aware construction algorithm
- Our designed index delivered significantly higher throughput and lower memory usage than prior solutions on metadata-indexing workloads such as LSM-tree compaction and DNS filtering.

Fast and Space-efficient Data Structure for Exact Multi-Set Membership Queries

February 2025 - December 2025

- Designed an efficient multi-set membership query algorithm, which outperforms all related work in query throughput, memory cost and query accuracy.
- The algorithm is a promising primitive in a wide range of network and database applications, including network traffic management and routing, distributed data storage/caching, security and privacy-preserving systems, and a variety of data mining tasks such as customer purchase analysis.

End-to-end system designed to consolidate co-existing NFs on heterogeneous devices

September 2023 - January 2025

- Designed an end-to-end system that takes desired NFs, device specifications, and user-defined objectives as input and automatically generates consolidated algorithms with a deployment plan, exposed through a curated NF library that makes it accessible to non-experts.
- Experimental results show the system produces resource-efficient consolidated algorithms for co-existing NFs.

ACADEMIC ACHIEVEMENTS AND RECOGNITION

- Outstanding Student Award of East China Normal University November 2022
- Outstanding Student Award of East China Normal University November 2021
- The Second Prize in the 17th China Graduate Mathematical Contest in Modelling November 2020

TEACHING EXPERIENCE

Teaching assistant, CSE 2050: Data Structures and Object-Oriented Design, University of Connecticut, Fall 2024, Spring 2025 and Spring 2026.